

Practical Implementation of an OFDM Based Active Radar System for Target Classification



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Declaration

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Date

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Acknowledgements

At the start of my engineering degree, a friend of my parents, who happens to be an engineer himself, heard that I had been accepted into the electrical engineering faculty at UCT. I asked him for any advice that he might have for me, anything that could potentially help me through the four years that lay ahead of me. He said one thing to me, ‘Make a group of close engineering friends, stick together, and you’ll get each other through’. This was some of the best advice that I have ever received. To everyone in the engineering friend group, getting to know everyone over the last few years has been one of the coolest experiences in my life. Thank you for all the memories made. I cannot wait to see what interesting things everyone gets up to in the coming years.

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Lastly, my family. Mum, Dad, Nic, and Mike, without your support, none of this would be possible. Mum and Dad, as I have grown older and hopefully slightly wiser, my appreciation towards your decisions to prioritise the education of us three brothers has grown exponentially. Thank you for all the support over the years, I really appreciate it.

Abstract

Every day without fail, all around the world, there are thousands of weather balloons that are launched into the atmosphere for meteorological observation and research applications. These weather balloons are all comprised of a latex balloon that is filled with helium, and a radiosonde unit that hangs below the balloon itself. This radiosonde module plays a very important role in the measurement of atmospheric conditions within which the balloon resides, as it contains the numerous electrical components that give it the ability to measure characteristics of the atmosphere and the position in which the radiosonde finds itself in, as well as transmit those readings back to a ground station.

Currently, there are only a handful of companies around the world that design, manufacture, and sell radiosonde modules. These companies essentially have a monopoly on the entire market. One of the hindrances to companies/individuals designing and manufacturing their own radiosonde modules is the complicated process of transmitting the recorded information back to a ground station. Since the companies that manufacture the radiosonde modules have closed designs on all of their components, information is not widely available as to the exact inner workings of their designs for their transmission systems, other than the broad modulation schemes. It is therefore of paramount importance that alternative transmission schemes to the schemes that are employed by the few companies that manufacture radiosondes are explored if alternative designs to the mass-manufactured models are desired.

This report outlines research into an alternative transmission system based on the [Long Range \(LoRa\)](#) communication system. While LoRa is still a proprietary communication system, it is widely available due to its popularity among hobbyist communities, as well as its expanding footprint within the [Internet of Things \(IOT\)](#) community of devices. Reasons for popularity within these communities include the long-range transmission capabilities of the communication protocol, as well as its low cost, which makes it an excellent choice for the disposable radiosonde modules that are used for atmospheric observations.

A transmission and receiver system was designed, simulated, and built, based around the LoRa transmission protocol. Tests were then performed on the system to determine the optimal LoRa parameters for the usage of the protocol in radiosonde modules. From testing and simulation data, the best bandwidth was determined to be 62.5kHz, and the best spreading factor was determined to be 11.

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Acronyms

AR Active Radar

ASR Airport Surveillance RADAR

AT&T American Telephone and Telegraph Company

BER Bit Error Rate

DAB Digital Audio Broadcasting

FDM Frequency Division Multiplexing

GPS Global Positioning System

IOO Illuminators of Opportunity

IOT Internet of Things

LoRa Long Range

OFDM Orthogonal Frequency Division Multiplexing

PAPR Peak to Average Power Ratio

RADAR Radio Detection and Ranging

Radcom RADAR Communications

SDR Software Defined Radio

TDM Time Division Multiplexing

TDMA Time Division Multiple Access

UCT University of Cape Town

USRP Universal Software Radio Peripheral

Chapter 1

Introduction

This report outlines the design and implementation of a single pulse classification [Radio Detection and Ranging \(RADAR\)](#) that uses an [Orthogonal Frequency Division Multiplexing \(OFDM\)](#) based signal with a flat top spectra. A literature review of RADAR and OFDM is presented, followed by the literature concerned with using OFDM signals in classification RADARs. Next, the design of the surrounding system to transmit and receive data, process the data, and analyse the results is presented. Following this, an experimental design is layed out and explained in detail. Next a simulation is performed and then the real world testing is performed. The results and comparison of the simulation and real world experiment are then discussed.

1.1 Background

RADAR technology has unlocked the ability for humans to be able to see the world around us with a new perspective. This ability to be able to see objects and more specifically, to see information such as position and velocity about these objects has helped with the development of devices and services that are of paramount importance to society. All the current use cases of RADAR do however lack the ability to determine exactly what object is being seen by the overall RADAR system. That is where the use of the OFDM waveform in a RADAR system becomes useful.

1.2 Problem Statement

Current RADAR technology works very well when determining the distance and direction of an object, however, it does not cope well with specific object classification tasks. One of the reasons for this is that current RADAR utilize either a pulse, continuous wave, or chirp waveform, which does not allow for object detection through statistical analysis due to the simple nature of the waveform. This is a drawback of using RADAR systems, since if object classification is desired, then a combination of radar and some other technology such as cameras would have to be used which increases the cost and complexity of the system as a whole.

1.3 Objectives

The goal of this project is to build a RADAR system that can simultaneously act as a normal RADAR which can determine the distance and direction of an object, as well as classify the specific object that it is looking at when compared to other example objects. The aim is to implement an OFDM

based waveform into a RADAR system that would give the flexibility to analyze the returned signal statistically and determine what exactly the object is. The goal is to make a non-realtime system, that can distinguish between three different objects.

1.4 System Requirements

The RADAR project has the following requirements:

1.4.1 Hardware Requirements

The main hardware for the project should be widely available computer scientific equipment. The availability of computer equipment is generally larger than most of the other required equipment such as the [Software Defined Radio \(SDR\)](#) equipment. The SDR equipment is much less widely available due to the high price of the equipment itself. This is fine however, since the [University of Cape Town \(UCT\)](#) has in their control multiple [Universal Software Radio Peripheral \(USRP\)](#) N210 devices designed and manufactured by Ettus research which will be perfect for my project.

test 1234

1. Be able to operate as a normal RADAR. This entails determining range, azimuth, and elevation.
2. Be able to classify between three separate reflectors/scatterers.

1.5 Scope and Limitations

The scope of this project has been set to not include design of RADAR hardware specifically. This is because design of RADAR hardware is a complex task that would probably encompass an entire masters project in and of itself. Therefore, the RADAR hardware portion of the project will be handled by a [SDR](#). This extrapolates the complex RADAR hardware design aspect away from the project and allows me to focus on the data collection and processing aspect of the project.

Chapter 2

Literature Review

In this chapter, I aim to explore the existing literature that delves into the fundamentals of the various elements involved within this project, starting with [RADAR](#), and more specifically, [Active Radar \(AR\)](#). Then I delve into [OFDM](#), followed by [SDR](#) and its working and uses within the world of RADAR. Then, I look into OFDM based AR, and finally, I look into OFDM based classification RADAR. My aim for this review of the literature is to add context to the broader paper, and hopefully give you, the reader, a more in depth understanding of the core concepts one has to understand to fully grasp the overall workings of the project. Especially as we move towards the latter parts of the project, where classification is involved, it is important to grasp the fundamentals.

I will separate the literature review into distinct sections, where I will explore the history behind each concept and the theory involved within. I will also explore and analyze the advantages and disadvantages of each topic and then finally, provide a critique of the literature towards the end.

2.1 Classical RADAR

2.1.1 Brief History

The fundamental idea behind RADAR systems and the technology itself were first theorized during the classical electromagnetic radiation experiments of the 1880s performed by Heinrich Hertz [2]. These experiments set out to prove the previous theoretical work achieved by James Clerk Maxwell. Maxwell came up with the formulae that described electromagnetic fields, and he hypothesized that this formula would apply to both light and radio waves since they were both examples of electromagnetic radiation just at vastly different frequencies. This has obviously since been proved, with visible light roughly falling into the 400THz to 700THz frequency range. The work done by Maxwell led to the discovery that electromagnetic waves can be reflected by metallic objects and refracted by a dielectric medium, just like light can [2]. This work did not go unnoticed and a patent for an ‘an obstacle detector and ship navigation device’ was issued in 1904 to Christian Hülsmeyer, a German engineer [2]. Work on RADAR then slowed down due to a lack of interest from governmental authorities until the early 1930s, when the development of long-range bombers prompted the major powers of the world to develop technologies which could detect these hostile aircraft, which would be useful for the coming second world war.

2.1.2 RADAR Workings

At its core, a RADAR is defined as an instrument that can detect surrounding objects using radio waves. To achieve this, there are multiple different setups of RADAR systems, which will be discussed later. These RADAR setups however all rely on the following fundamental laws of physics which are listed below [3]:

1. **Reflection of electromagnetic waves:** This rule describes how a radio wave will be reflected back towards a transmitter of said radio wave if the object is placed directly in front of the transmitter. This reflection indicates that the object is in front of the transmitter.
2. **Electromagnetic energy travels through air at a constant speed:** This rule describes how the electromagnetic waves transmitted by an antenna propagate through the transmission medium, usually air for our earth's use cases, at roughly the speed of light, c . This velocity c equates to $299,792,458 \text{ m.s}^{-1}$
3. **Electromagnetic energy normally travels through space in a straight line:** This describes how the energy transmitted by an antenna in a radar system propagates through space in a straight line. This is important because it means that the energy of the RADAR system can be directed in a specific direction by the antenna system, which means that the reflecting object's direction (azimuth and elevation) can be determined.

In a classic pulsed RADAR, essentially the most simple kind of RADAR that is used today, by using the above-mentioned physical properties, the distance, azimuth, and elevation of the object that is being observed by the radar system can be determined. The best example of this sort of RADAR system is an airport RADAR. An [Airport Surveillance RADAR \(ASR\)](#) is used at airports to determine the presence and position of aircraft in the vicinity of an airport. Typically, ASR systems are capable of reliably detecting and tracking aircraft at altitudes below 25,000 feet (7,620 meters) and within 40 to 60 nautical miles (75 to 110 km) of the airport [4].

Classical RADAR systems can be separated into a few different categories [1]. Broadly, you get primary and secondary RADARs that are separated into imaging and non-imaging RADARs. An imaging RADAR is one which desires to draw an image, such as a weather RADAR, and a non-imaging RADAR is one which does not draw an image but rather infers information about a target based on processing techniques. Primary RADARs rely on echos from a target that come from it's own transmitted pulse, whereas a secondary radar relies on a RADAR response signal from a target. Sometimes, the ASR systems mentioned previously have secondary radar systems built-in which allow the aircraft being detected to broadcast vital information to aid with air traffic control. This is usually only possible for 'friendly' targets. The various categorizations of RADAR types can be seen in figure 2.1 below:

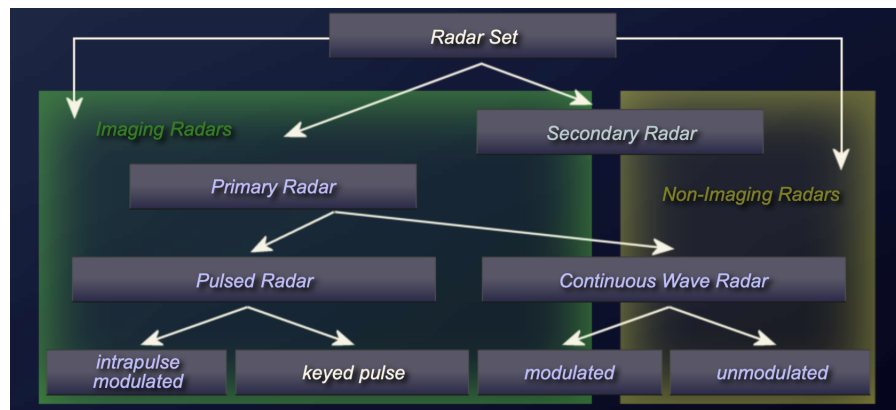


Figure 2.1: Classification of Radar Systems [1]

Note to self, I don't think I like this image

2.1.3 Advantages and Disadvantages of RADAR

Since RADAR utilizes the electromagnetic spectrum, there are distinct advantages concerning the use of RADAR systems [3]. The main advantage is that, unlike humans, RADAR systems can operate day and night, 24/7, over far distances, in good or bad weather. Compared to human eyes, this is obviously a lot better. Some other advantages of classic RADAR is that it can generally 'see' through insulators, and it can also distinguish the difference between fixed and moving targets [5]. Some disadvantages however, are that classical RADAR systems generally struggle to resolve targets that are very close to the transmitter, and they cannot penetrate water very well [5]. Classical RADAR systems also fail to distinguish between shapes of object that they are looking at.

2.2 OFDM

2.2.1 Brief History

The general working principles, and the mathematics required for OFDM were originally described by Robert Chang in 1966 while he was working at Bell Labs [6]. While the idea of OFDM was smart, it didn't become popular until sufficiently cheap microprocessors entered the market [6]. OFDM was never implemented in commercial communication channels until the appearance of mass market applications coincided with the availability of efficient software and electronic technologies [7].

2.2.1.1 Previous Transmission Techniques

Before OFDM there was [Frequency Division Multiplexing \(FDM\)](#). FDM system innovation was driven out of necessity due to telegraph companies hoping to increase the amount of users that could utilize their transmission lines to increase profits from their systems [7]. Transmission companies were in search of a method to have multiple messages being sent on the same transmission line at simultaneously. Other methods had been used such as [Time Division Multiplexing \(TDM\)](#), and [Time Division Multiple Access \(TDMA\)](#). These methods were however limited by burst speed which was itself limited by inter-symbol interference. Around the 1920s, FDM became the main carrier system for telephone companies. The

first major company to deploy the system was [American Telephone and Telegraph Company \(AT&T\)](#). This initial deployment of the system used 8kHz subchannels with 4kHz allocated to each channel. This form of transmission system later developed into systems that had more subcarrier channels with less subcarrier spacing. There were however some drawbacks to FDM, which are highlighted below:

1. **Waste of frequency:** Due to the large subcarrier spacing, FDM wasted a lot of precious frequency spectrum to these so-called ‘guard’ bands on either side of the carrier. As the frequency spectrum became more crowded throughout the years, these guard bands became a liability due to demand for spectrum being too high, and the guard bands were not efficiently using the spectrum.
2. **Using different modulators for the various sub-channels:** In the old systems, each sub-channel had a separate modulator in the up-mixing process. This multiplication process became tedious once there were multiple sub-carriers.

To compensate for this inefficient use of the frequency spectrum, OFDM was introduced. What made OFDM different to FDM was that each subcarrier overlapped some of the other carriers in signal. This was able to be done because each carrier was spaced in such a way that it was orthogonal to the other carriers in the signal [7].

2.2.1.2 OFDM

As mentioned before, [OFDM](#) is a transmission method that uses the spectrum in a more efficient manner to transmit as much information as possible over a single transmission line. The transmission of information is commonly done using various modulation schemes such as QAM encoding that breaks information into bit chunks that are sent in parallel over multiple channels [8]. The OFDM transmission scheme builds upon FDM where multiple channels are created over a specific bandwidth, and bits of information are sent on each channel in parallel. The main difference between FDM and OFDM is therefore the orthogonality of the OFDM signal. I have created a crude example below in figure 2.2. In fig 1.2a, a crude mockup of an OFDM signal can be seen in the frequency domain. As can be seen, all the peaks of the OFDM waveform have zero interference with any other part of the signal. This is due to the various carriers being orthogonal to each other. Conversely, with the FDM waveform, the peaks have significant interference from the other channels since the channels are not spaced orthogonally in regard to each other.

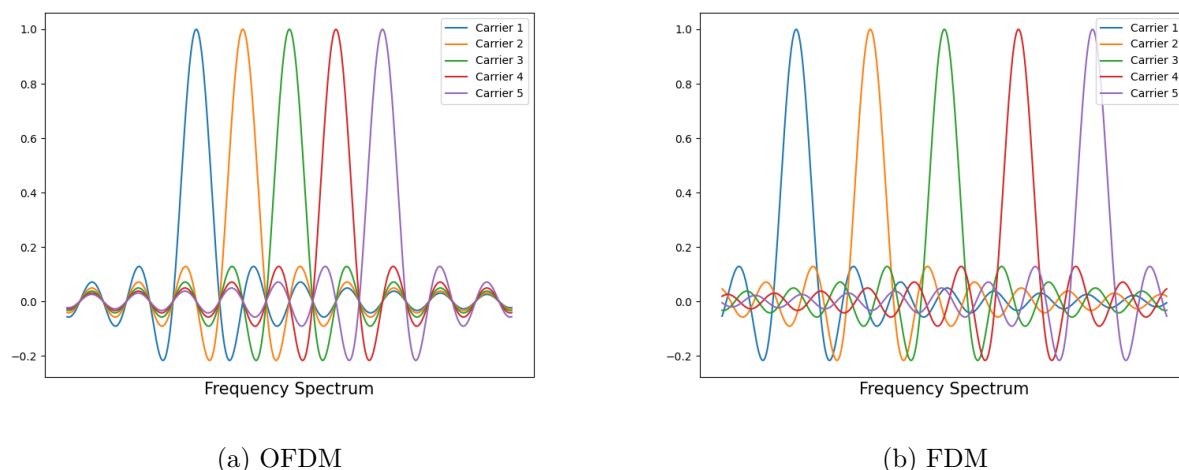


Figure 2.2: OFDM and FDM Waveforms

I want to touch on the idea of orthogonality. In practice, all it means is that in a signal with multiple sinusoidal carriers, each carrier has a frequency that is a whole number multiple of some fundamental frequency [6]. The time domain voltage signal of an OFDM signal can be described by equation 2.1 below:

$$v(t) = \sum_{k=0}^{N-1} X_k e^{\frac{j2\pi kt}{T}}, \quad 0 \leq t < T \quad (2.1)$$

N = Number of subcarriers

where X_k = Data symbols in the frequency domain

T = OFDM symbol time (s)

The subcarrier spacing of $\frac{1}{T}$ ensures carrier orthogonality over the OFDM frame, or symbol. In practice, real world implementations of OFDM signals will have vastly more complex components added on concerning error correction and data modulation techniques in general. The base signal however will still revolve around the idea of orthogonal carriers.

2.2.2 Advantages and Disadvantages of OFDM

As can be seen in the above sections, OFDM is a very useful and versatile transmission technique that is used in a variety of real world applications. While OFDM is a widely used and very good transmission technique, it does come with some pros and cons [9]. The biggest pro for OFDM, particularly with its use within the telecommunications industry, is the spectral efficiency that can be achieved. Real world OFDM systems are also resistant to multipath fading due to the cyclic prefix that these systems implement. Modern OFDM systems can allow for easy equalization with embedded pilot symbols. Equalization is much harder to achieve in a single carrier system. Some cons of OFDM systems however include the following. Firstly, an OFDM transmission can have a high [Peak to Average Power Ratio](#)

(PAPR). This is due to an OFDM signal's noise-like amplitude with a large dynamic range [9]. Another problem is that OFDM systems are susceptible to carrier frequency offset due to local oscillators not being able to perfectly operate at the same frequency on the transmitter and receiver.

2.3 OFDM based RADAR

2.3.1 Brief History of OFDM based RADAR

2.3.2 Implementation of OFDM based RADAR

There is an ever evolving area of RADAR research that involves the combination of RADAR systems and telecommunication systems. This is colloquially known as [RADAR Communications \(Radcom\)](#). Radcom aims to combine radar and telecommunications because this could potentially lead to system that are more spectrally efficient, and lightweight. In essence, one could imagine a RADAR system that can communicate with other base stations, as well as performing usual RADAR tasks, all from a single system. An attempt at creating a system like this was attempted by Dr Stephen Paine and Grant Norrie in [10]. They set out to create a RADAR system that utilizes the previously mentioned OFDM signal structure in a radcom setup. They wanted to base the system around an OFDM signal because of the ever-increasing popularity of OFDM signals in communication networks. They mention that communication signals are often used in passive RADARs, by exploiting [Illuminators of Opportunity \(IOO\)](#). An IOO is a transmitter that is not intended for RADAR purposes, but rather other applications, with the most popular being telecommunications purposes. They pose the question, if a passive RADAR system can use existing communication signals to detect objects, why can an active system not do the same?

They explored the use of the OFDM based [Digital Audio Broadcasting \(DAB\)](#) modulation scheme as the backbone of the system. Part of the reason for this, is the fact that DAB is a very popular and well-defined. They designed and documented an entire toolchain that consisted of the following parts.

1. Signal Generation Chain: This entailed the building of the signal that was to be transmitted by their receiver.
2. Hardware Testbed: This was the physical hardware that was used to implement the theoretical in the real world. This consisted of dual Ettus USRP N210 SDRs.
3. Radar Processing Chain: Everything to do with processing of the signal on the reception side of things to verify the operation of the system as a RADAR. The main objective of this toolchain was to generate a range-doppler graph to visualize the output of the overall RADAR system.
4. Communications Processing Chain: This toolchain was in place to verify and perform the necessary operations to facilitate a working communication link between the transmitter and the receiver.

The SDR platform had specific configuration characteristics that were used for the tests performed by [10]. These settings are summarized in table 2.1 below:

Parameter	Value
Tx Gain	0 dB
Rx Gain	0 dB
Tx Bandwidth	1.1 MHz
Rx Bandwidth	1.1 MHz
Sampling Rate	5 MHz
Carrier Frequency	2.45 GHz

Table 2.1: SDR Toolchain Parameters

Multiple tests were performed to test the capability of the system. These tests were a communication test, a static range test, and a range-doppler test, which involved a moving target. The communication test was fairly basic, and it aimed to eliminate antenna/system feed through, and then measured the [Bit Error Rate \(BER\)](#) upon transmission from the transmit antenna to the reception antenna from a distance of 5 meters away. The BER was confirmed to be 0% and then the test was deemed to be a success.

Then a static range test was performed with an object 100m away from the RADAR. After pulse cancellation with adjusted dynamic range, and clutter map removal, a range-doppler map was generated as shown below in figure 2.3:

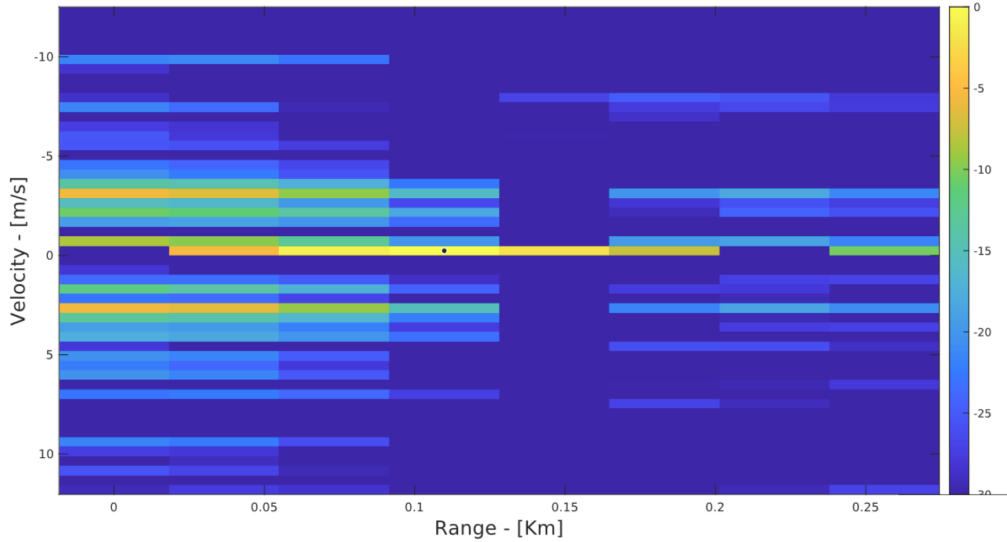


Figure 2.3: Range-doppler map of object at 100m

This figure showed that the object was detectable at 100m, specifically, the object could be seen in the 109m range bin. Lastly, a moving target test was performed, which was able to successfully detect a moving car. The ground truth of the moving object was as follows, it was 30m away from the RADAR and was moving at a velocity of 7.2 m.s^{-1} and 7.0 m.s^{-1} as measured by a [Global Positioning System \(GPS\)](#) and the odometer respectively. The measured range and velocity was determined to be 36m and -7.4 m.s^{-1} . The negative velocity is due to the object moving away from the RADAR system.

In [\[11\]](#), Gabriel Lellouch

2.3.3 Advantages and Disadvantages of OFDM based RADAR

2.4 Critique of Literature

Notes to self

...In Stephen Paine and Norrie paper, the communication test is quite trivial...

Chapter 3

System Design

[3]

3.1 Requirement and Specification Analysis

3.1.1 System Requirements

3.1.2 System Specifications

3.2 Modules and Sub-modules

3.3 Transmission Module Component Selection

3.4 Receive Module Component Selection

3.5 Hardware Interface Design

3.6 Software Design

3.7 PCB Design

Chapter 4

Validation and Simulation

[3]

4.1 Validation

4.1.1 Transmitter Validation

4.1.1.1 Power Module

4.1.1.2 Microcontroller

4.1.1.3 Sensor Modules

4.1.1.4 LoRa module and Display and Control Module

4.1.2 Receiver Validation

4.2 Simulation

Chapter 5

Methodology

[3]

5.1 Research Design and Approach

5.2 Data Collection

5.3 Shorter Range Tests

5.4 Long-Range Test

Chapter 6

Results

This section outlines the results of the experiments performed that are outlined in section 5. Discussion and analysis of results will not form part of this chapter, but rather the raw data captured will be presented for all the important tests that took place. [3]

6.1 Same Room Test

6.2 100m Test

6.3 300m Test

6.4 500m Test

6.5 20km Test

6.6 Average RSSI Values with Distance

Chapter 7

Discussion

[3]

7.1 RSSI Values with Distance

7.2 Overall Testing Performance

7.3 Determining the Optimal Packet Structure

7.4 Determining the Optimal LoRa Parameters

Chapter 8

Conclusion and Recommendations

[3]

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Chapter 9

Appendix

[3]

9.1 Full Transmitter Circuit Schematic

9.2 Full Receiver Circuit Schematic

9.3 Long Range Test Interference

9.4 GitHub Link

9.5 Ethics Clearance